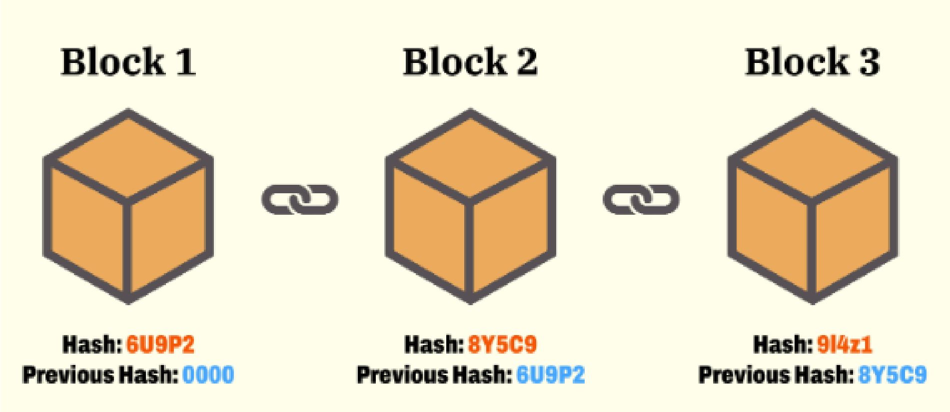
Blockchain
Two

Revision From Week One & Two

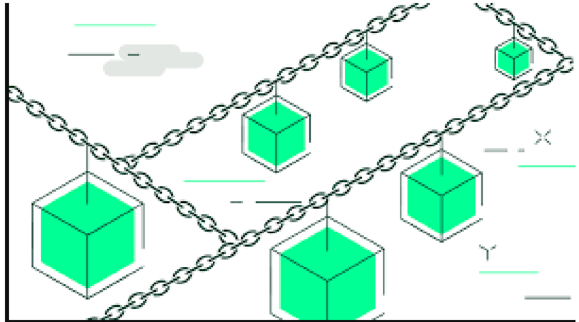
- BTC
- Who and Where
- Components / Technological
- Historic
- Some Stories – Growth and Pop Culture
- Miners & Mining
- Difficulty
- Cryptography



- What is Blockchain
- Block – Each Transaction is Taken from a Pool of Transactions and Formed into a Block of Transactions.

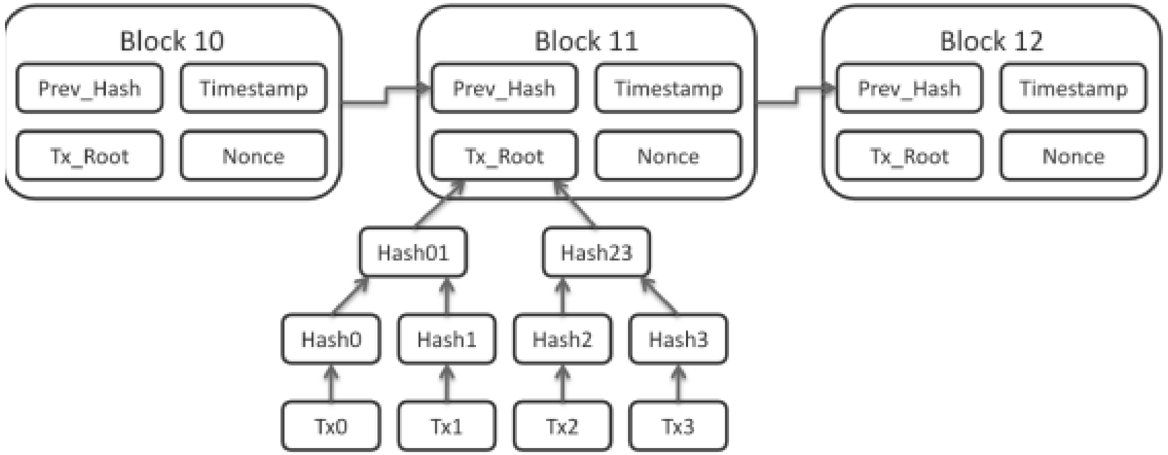
- The Chain – These are Cryptographic Functions which link each block together

- Blockchain
- Blockchain Should or Can , be considered as a Sub-Set of DLT
- Distributed Ledger Technology

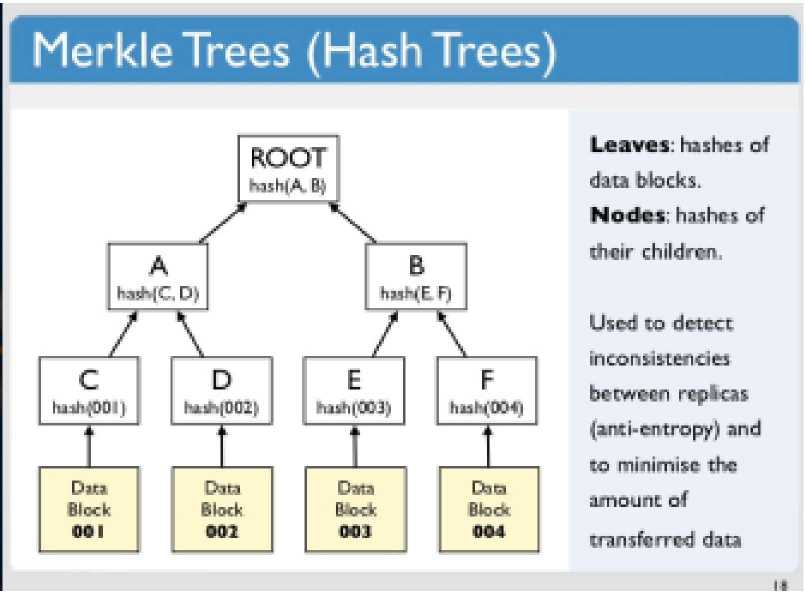


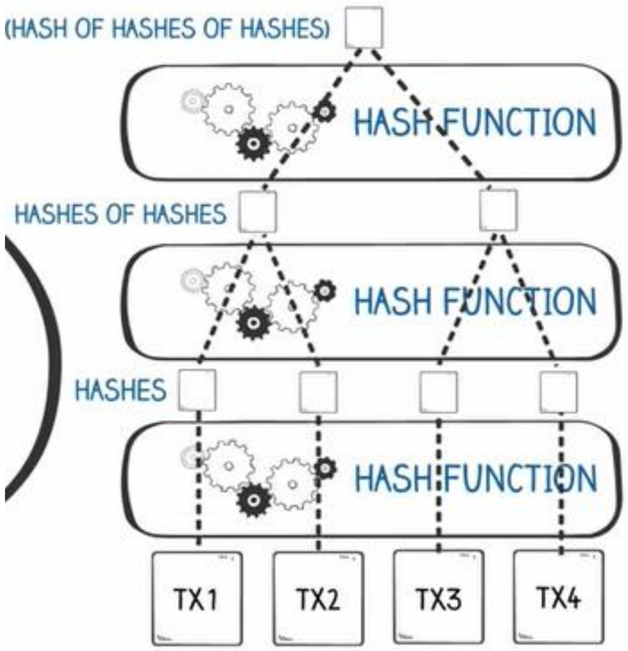
Cryptographic Functions Linking Blocks

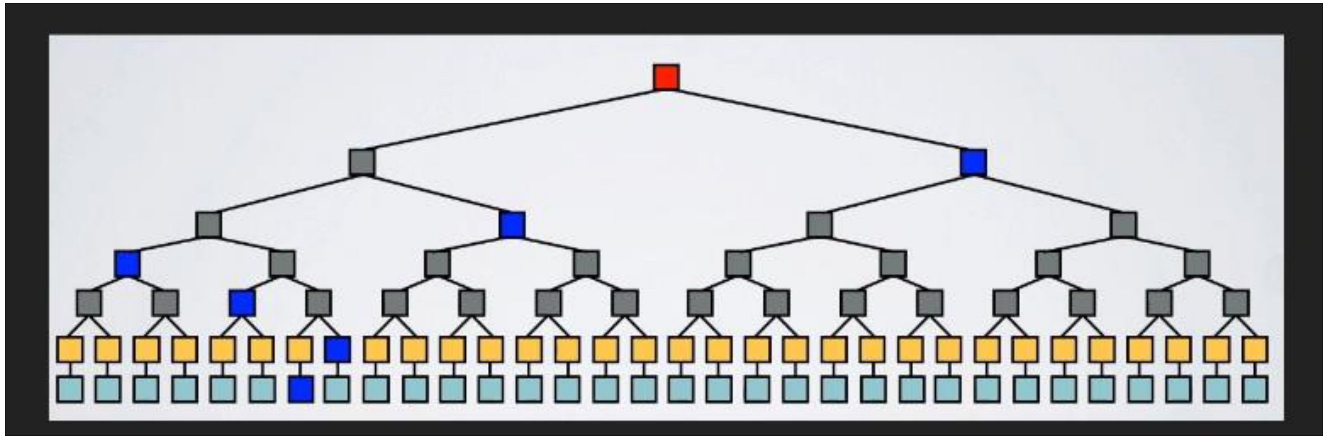
- The Block and Header
- Within Each Block which is Mined to the Blockchain
- We have a typical composition / or Cross-section
- This is known as the Block Header :
- What do you See ?



Dr Merkle and His Tree





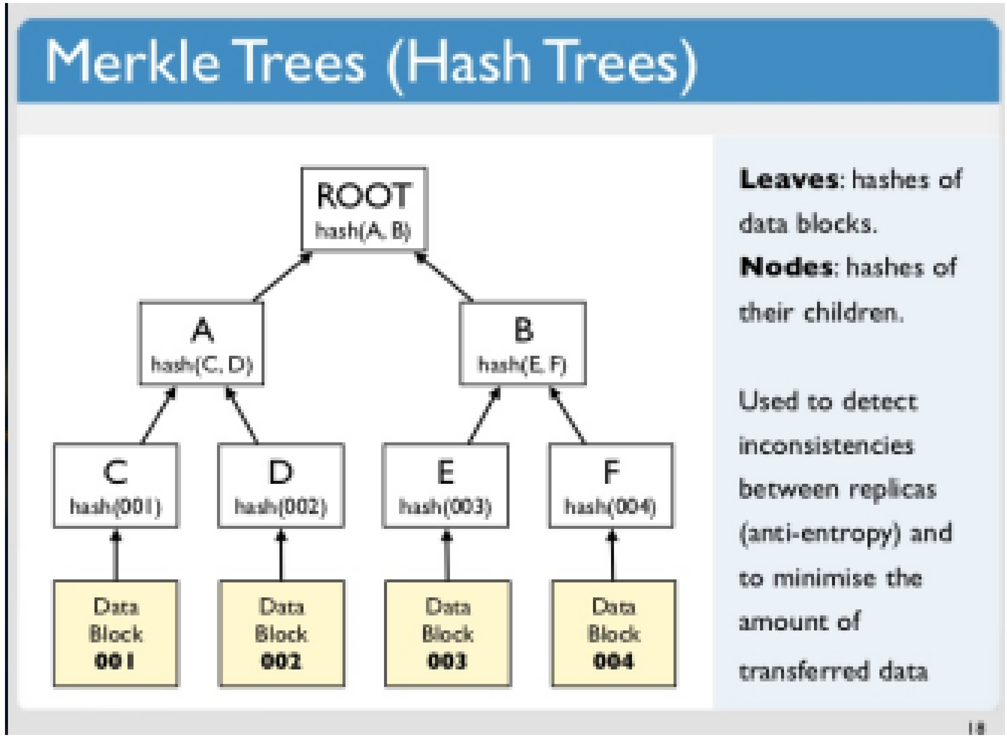


Merkle Tree

- Each block in the Bitcoin blockchain contains a summary of all the transactions in the block in the form of a Merkle tree. Also known as a binary hash tree, it is a data structure used for efficiently summarising and verifying the integrity of large datasets.

- Double SHA-256 algorithm applied to tree to create the block hash, which is the primary identifier of a block which any node can independently derive by simply hashing the block header.

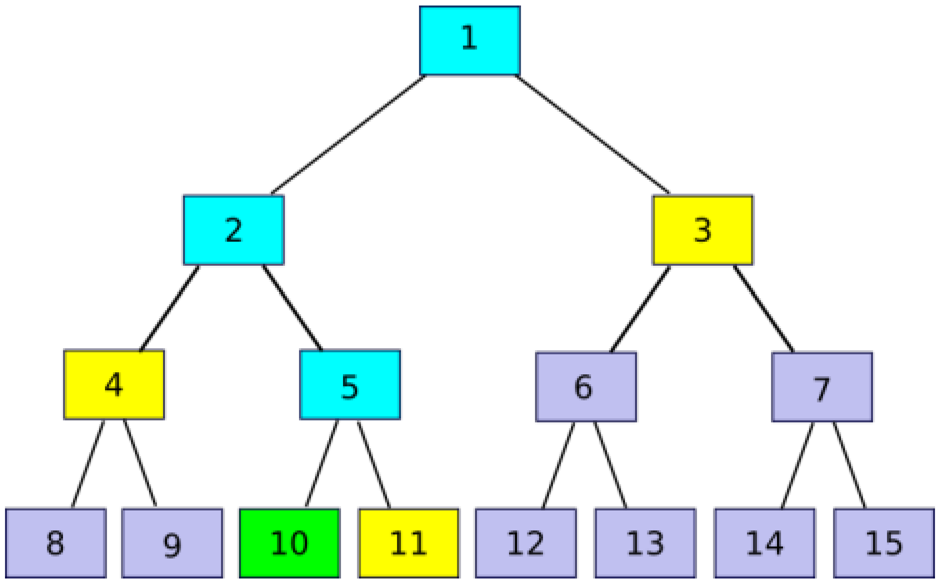
- The block height signals the position of the block in the chain. While it is true that block hash will always uniquely identify a block, the same isn't true about block height.



- Used extensively by SPV (Simple Payment Verification) wallets such as Electrum who do not require you to download the full blockchain

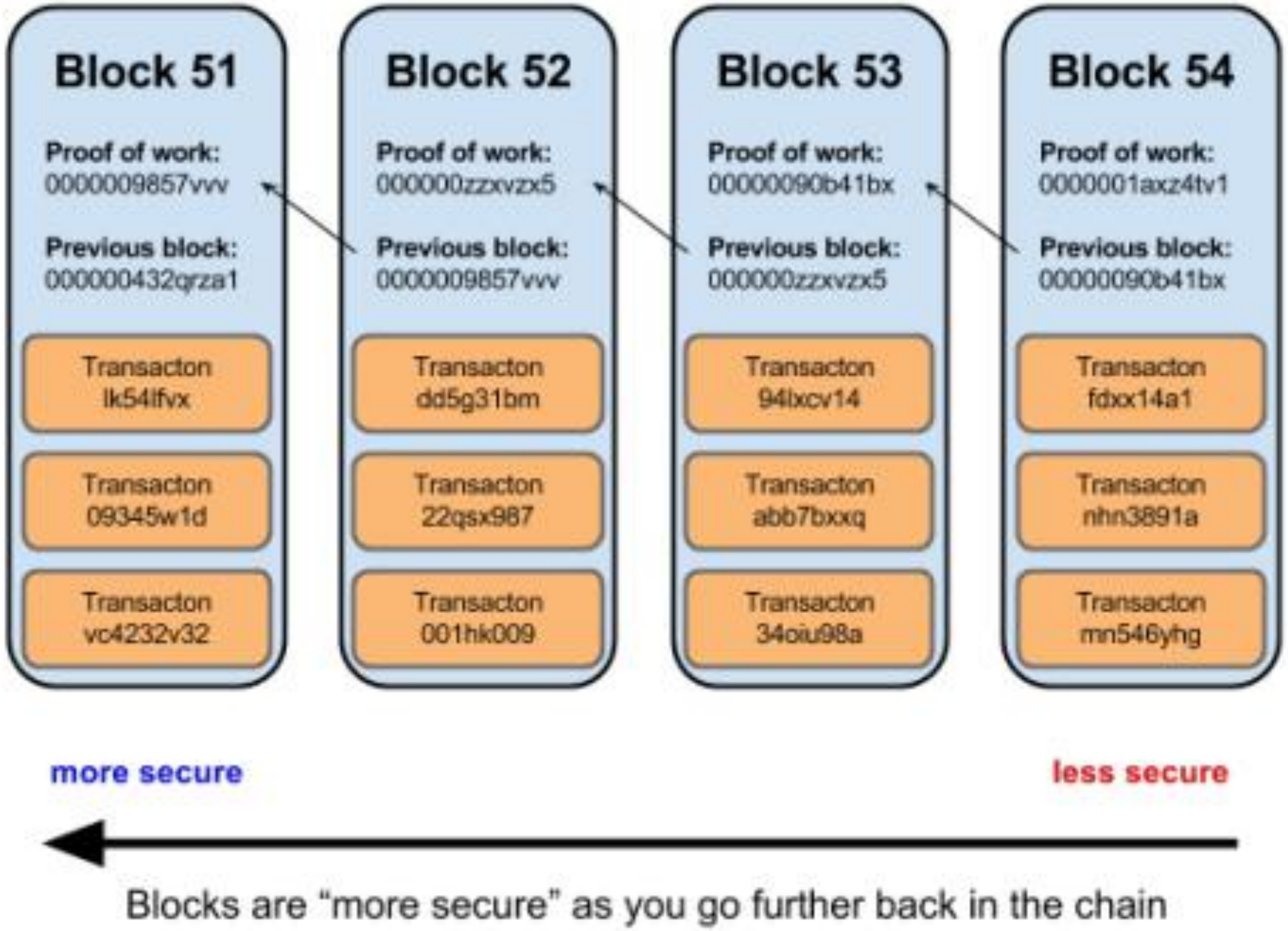
Merkle Proof

- Ralph Merkle
- Patented the Merkle Tree in 1979
- The Top Hash called the Merkle Root
- Merkle Trees can be used to provide
  - Merkle Proofs
  - Proof that a piece of data is part of the Data set represented by the Merkle Proof



Immutability

- Anders Blockchain Example



Difficulty in BTC

- Difficulty is adjusted every 2016 Blocks (2 Weeks)
- Difficulty is intended to keep the Block interval at about 10 Mins
- A Valid Block has a hash value which is lower that the target difficulty

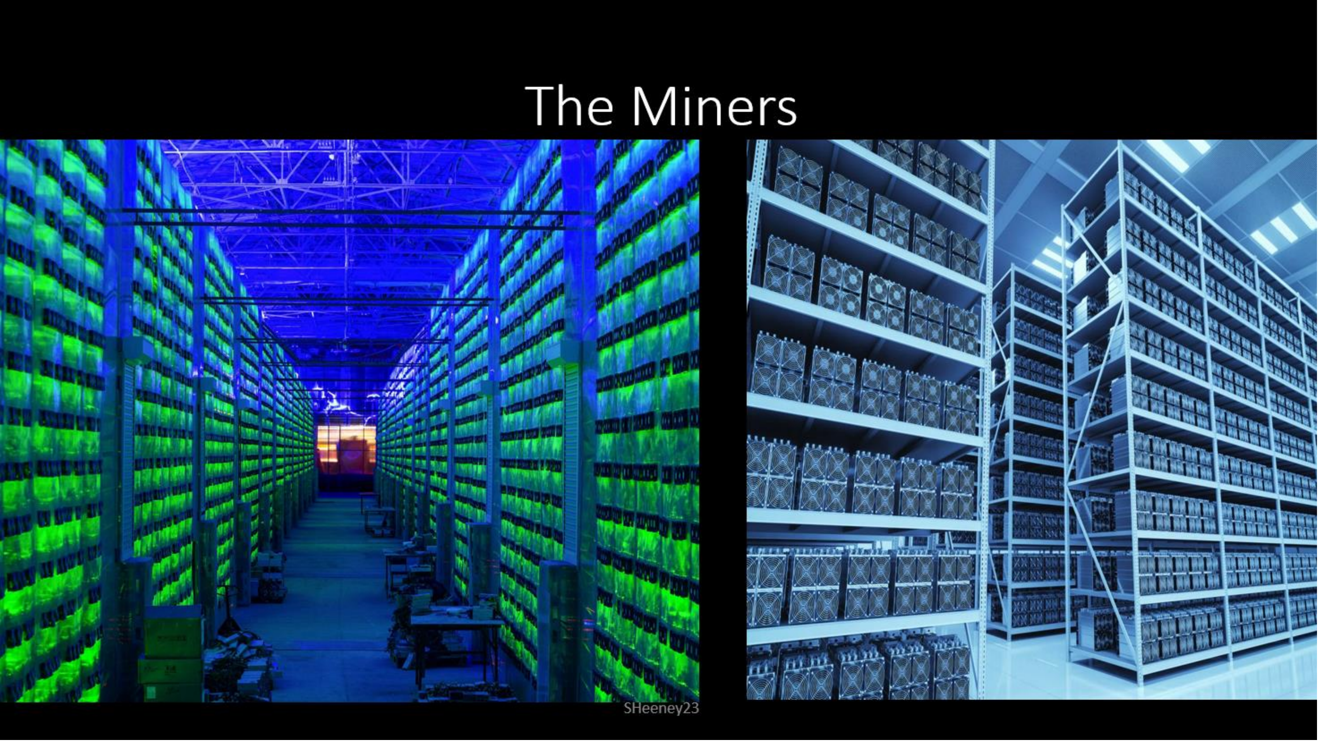
000000000000000000cd9ce061de1770365b0841980cde2d8f79502385beb8
9c

POW Proof of Work

- How does it provided Immutability ?
- How does the Block Hash provide the work has been completed ?

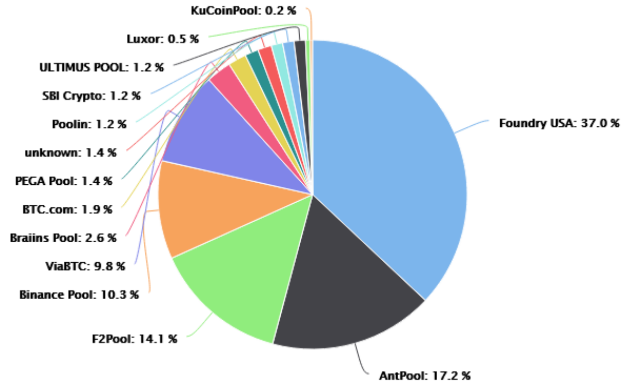
- Why is 51% an important figure ?

- Example ::
  - Bitcoin Network Hash Rate: 138 000 000 000 000 000 000 H/s
  - Assume 10c kWh cost
  - Antminer S9
    - 14 THs
    - 1372W Power Consumption
    - $1700 ex VAT



https://www.asicminervalue.com/

https://btc.com/stats/pool



https://www.asicminervalue.com/miners/bitmain/antminer-s9-13-5th

ASIC Miner Value | Miners | Opportunities | Directories | Help | BTC = $22,954.35

Bitmain
Antminer S9 (13.5Th)

BITMAIN

Profitability

Period	/day	/month	/year
Income	$1.17	$35.25	$422.98
Electricity	-$3.81	-$114.31	-$1,371.69
Profit	-$2.64	-$79.06	-$948.70

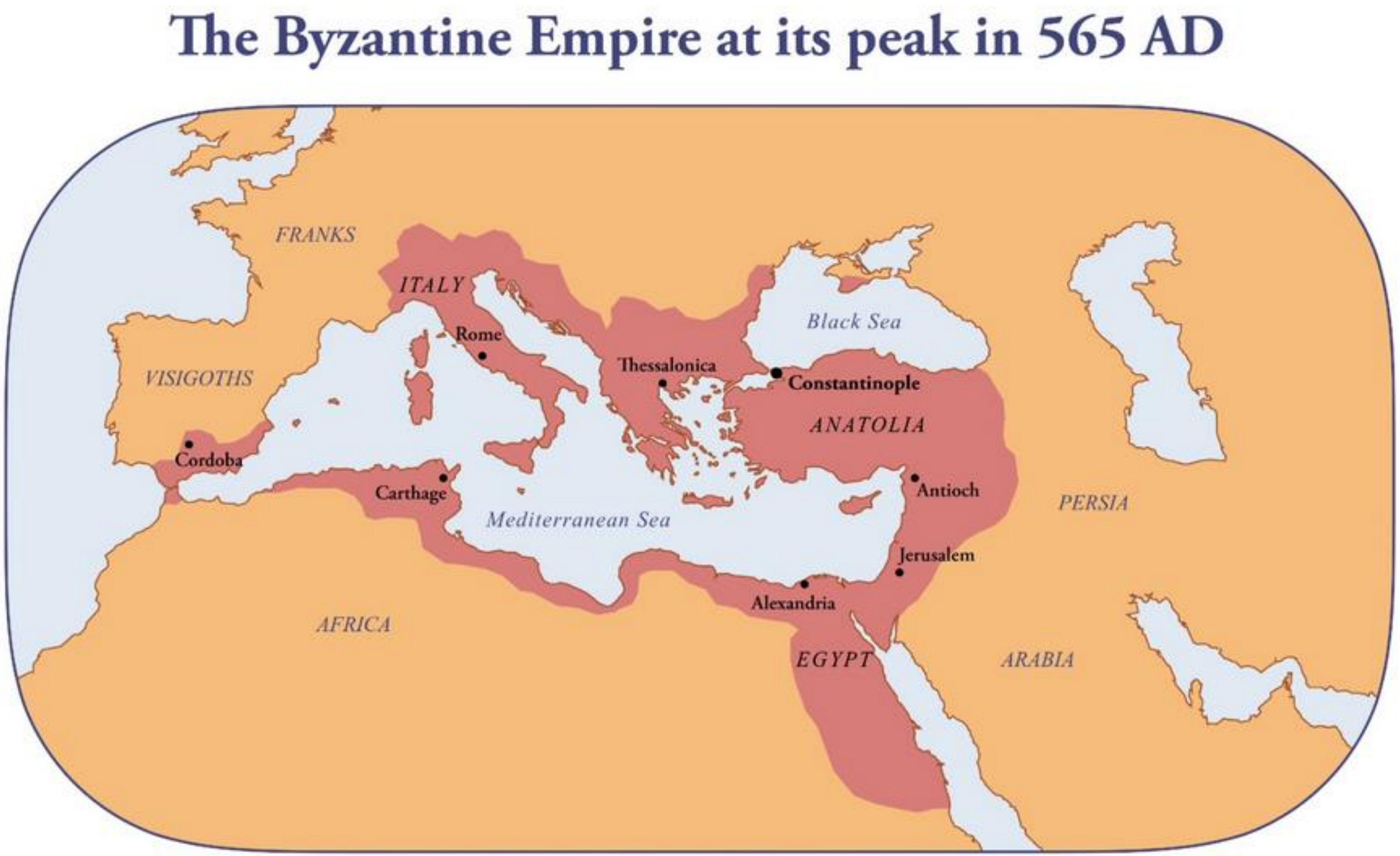
Algorithms

Algorythm	Hashrate	Consumption	Efficiency	Profitability
SHA-256	13.5Th/s ±5%	1323W ±10%	0.098j/Gh	-$2.64 /day

Description

Model **Antminer S9 (13.5Th)** from **Bitmain** mining **SHA-256 algorithm** with a maximum hashrate of **13.5Th/s** for a power consumption of **1323W.**

Byzantine Fault Tolerance

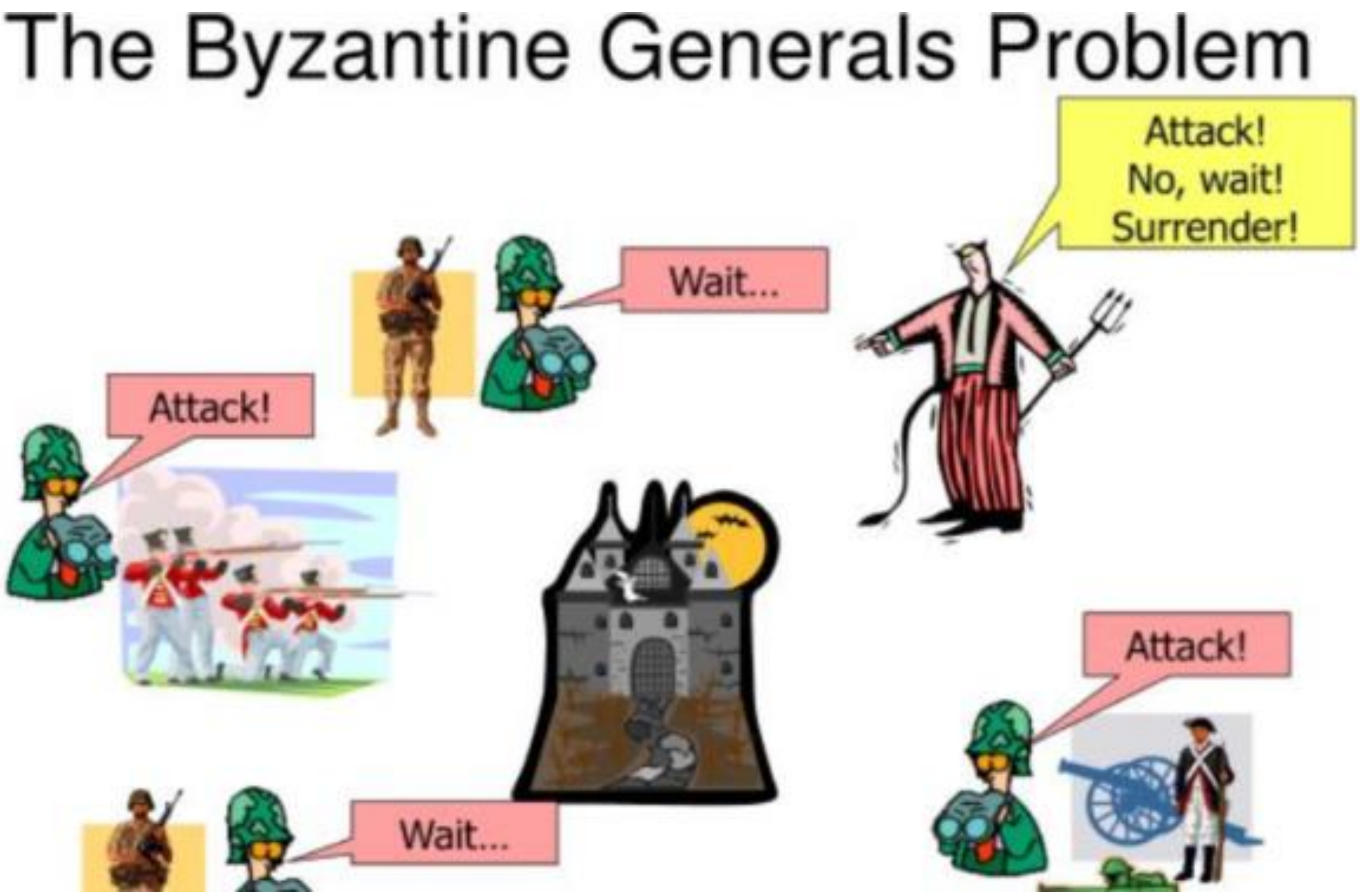



The Fall of Constantinople

The Byzantine Generals Problem

- Lamport , Shostak, Pease (1982)
- All Generals Must Co-Ordinate the attack
- Attacks are co-ordinated by messengers
- Messengers must travel through enemy territory
- Messengers can forge messages
- Generals can send multiple messengers
- An uncoordinated attack will fail

Bitcoin solves the Byzantine Fault Tolerance problem by using Proof of Work, which increases the cost/time to generate a message compared to propagation time, which ensures a consensus is reached around longest valid chain


The Byzantine Generals Problem
Attack!
No, wait!
Surrender!
Wait...
Attack!
Attack!
Wait...

Important Points in Crypto History

- **1 st November 1998**

Wei Dai – b-money

- **1 st August 2002**

Adam Back – Hashcash

- **15 August 2004**

Hal Finney – Reusable Proof of Works

Hal Finney - Cyperphunks announcement

- **December 27 2008:**

Nick Szabo – Bit gold

- **February 11 2009:**

Satoshi – P2P foundation

I am fascinated by Tim May's crypto-anarchy. Unlike the communities traditionally associated with the word "anarchy", in a crypto-anarchy the government is not temporarily destroyed but permanently forbidden and permanently unnecessary. It's a community where the threat of violence is impotent because violence is impossible, and violence is impossible because its participants cannot be linked to their true names or physical locations. “

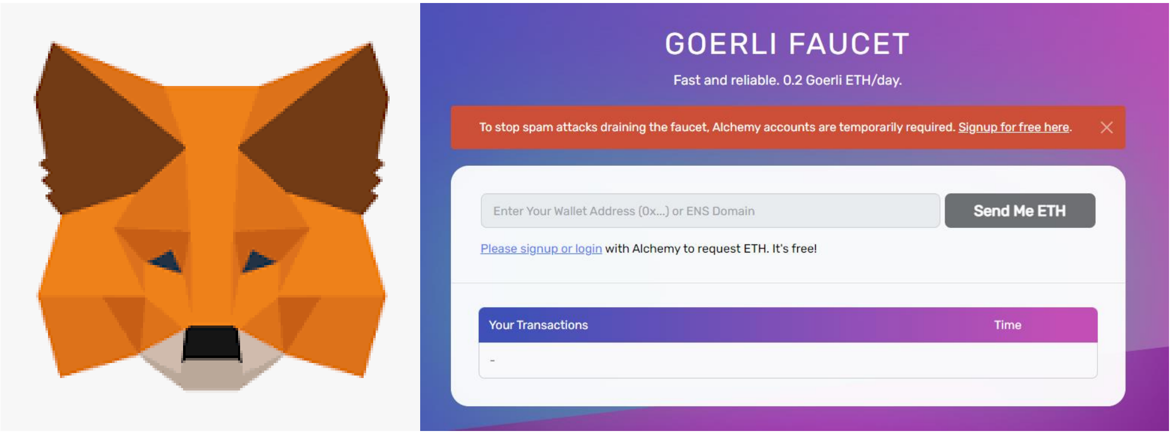
(Dai, 1998)


ethereum
HOMESTEAD RELEASE
BLOCKCHAIN APP PLATFORM

Notes on MetaMask & Faucets

- Overview:
- https://ethereum.org/en/developers/docs/networks/

- Explorers:
- https://goerli.etherscan.io
- https://sepolia.etherscan.io/



Faucets
Sepolia
https://sepolia-faucet.pk910.de/

Gorli:
https://fauceth.komputing.org/?address=0x0b97Bc397357e6b95435a9b8B3D17FbcD528173E

Alchemy
https://goerlifaucet.com/

NotGreat:
https://faucet.quicknode.com/ethereum/goerli/?transactionHash=0x72a42624ccb50d5f2c4fe637babb1adb4db147001b536412ae01340074de3961

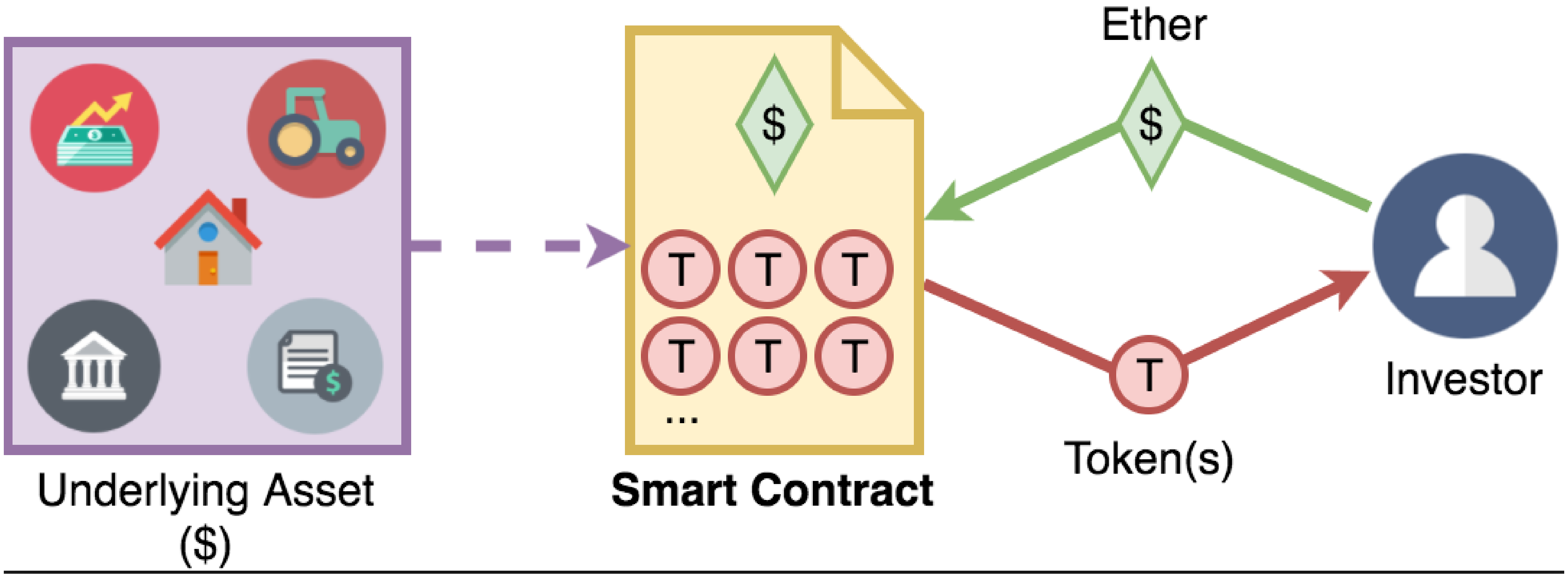
Other_MyCrypto_Dashboard: Only Supports Goerli
https://app.mycrypto.com/dashboard

```
contract myToken is ERC20Interface {

    string public symbol;
    string public name;
    uint8 public decimals;
    uint public _totalSupply;
    address public tokenOwner;

    mapping(address => uint) private _balances;
    mapping(address => mapping(address => uint256)) private _allowances;

    constructor() public {
     tokenOwner = msg.sender;
     symbol="TOK";
     name="Fixed supply token";
     decimals=18;
     _totalSupply = 5 * 10**uint(decimals);
     _balances[tokenOwner] = _totalSupply;
     emit Transfer(address(0), tokenOwner, _totalSupply);
    }

    function totalSupply() public view override returns (uint256) {
     return _totalSupply - _balances[address(0)];
    }
```

Smart Contracts Prologue,

an event or act that leads to another.

- Defined in 1996 by Nick Szabo
- Computer protocol used to facilitate, verify or enforce a legal contract.
- The smart contract is the code itself, and is used to govern all aspects of a contract in order to free it from requiring a third party to confirm all aspects of the contract have been undertaken and satisfied (Decentralisation).
- Read the Docs : https://ethereum.org/en/developers/



Ether
$
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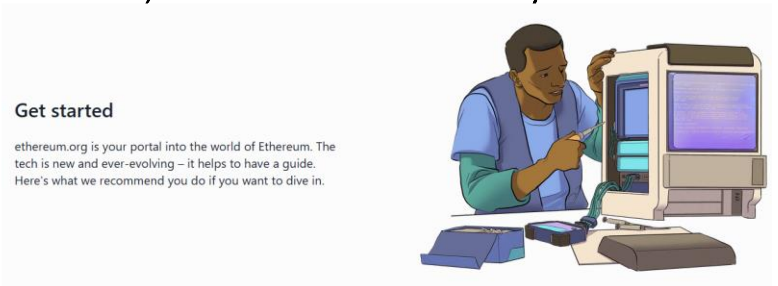
...
T
Investor
Token(s)
Underlying Asset ($)
Smart Contract

Electronic Contracts , but much more ! !

- Control / Code
- Hypothetically, your bank could arbitrarily add or remove money from your account.

- In a blockchain ecosystem, there should be no single source of control. The distributed nature of the network with the consensus mechanisms result in multiple parties performing checks, with breaches to the pre-conformed rules rejected by consensus
- Transparency
- Anyone can see and review the code i.e the contract
- Are there positives and negatives related to this point?

- Current implementation: Send a direct debit on the 5th of every month to Michael
- Possibilities with smart contracts : Send a direct debit on the 5th business day every month. If earnings for the previous month are less than 10k, then send 1k. If earnings are greater than this, send 1k plus 10% of the difference.
- Rainy day smart contract: Every day it is sunny, send 3.33% of my salary to my savings account. If it rains, transfer this back to my current account.



- Ethereum
- Implications of this:
- Decentralized: No controlling entity or individual * Discuss
- No Single point of failure
- Immutability * Discuss
- Corruption / Tamper proof – based on consensus , making censorship impossible
- Zero Downtime
- Difficult to control from a regulatory perspective

- DAO (Decentralized Autonomous Organisation) :
- Fully autonomous organisation with no single leader
- Run by a series of smart contracts designed to replace the rules and structure of a traditional organisation eliminating the need for centralized control
- Ownership represented by tokens, which act as token rights.

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    string public name;
    uint8 public decimals;
    uint public _totalSupply;
    address public tokenOwner;

    mapping(address => uint) private _balances;
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    constructor() public {
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